Arham Lodha
MATH 245A

Discussion 3
Due:

1. Measurable functions

Problem 1.1

(Level sets; Bass 5.1.) Let $(X, \mathcal{A})$ be a measurable space and let $f : X \to \mathbb{R}$. Suppose that

$$\{x : f(x) > q\} \in \mathcal{A} \quad (q \in \mathbb{Q}).$$

Prove that f is measurable. Deduce that if $f_n : X \to \mathbb{R}$ are measurable, then $\sup_n f_n$, $\inf_n f_n$, $\limsup_n f_n$, and $\liminf_n f_n$ are measurable (when allowed to take extended-real values).

> *Solution.*

Problem 1.2

(Where does a sequence converge?; Bass 5.4.) Let $f_n : X \to \mathbb{R}$ be measurable. Prove that

$$C = \left\{x : \lim_{n\to\infty} f_n(x) \text{ exists in } \mathbb{R}\right\}$$

is measurable.

> *Solution.*

Problem 1.3

Give an uncountable family of nonnegative measurable functions $\{f_\alpha\}_{\alpha\in I}$ such that

$$f(x) = \sup_{\alpha\in I} f_\alpha(x)$$

is finite everywhere but is not measurable.

> *Solution.*

2. Borel–Cantelli and convergence

Let $(X, \mathcal{A}, \mu)$ be a finite measure space, i.e. $\mu(X) < \infty$.

Problem 2.1

(The first Borel–Cantelli lemma.) Let $E_n \subseteq X$, $n = 1, 2, \cdots$ be a sequence of measurable sets. Prove that

$$\sum_{n=1}^{\infty} \mu(E_n) < \infty \implies \mu\left(\limsup_{n\to\infty} E_n\right) = 0.$$

Show by example that the converse is false.

Also show that if $\mu(E_n) \geq \delta > 0$ for every n and $\mu(X) < \infty$, then $\mu(\limsup E_n) \geq \delta$.

> *Solution.*

Problem 2.2

(An almost-everywhere convergent subsequence.) Suppose that $f_n, f : X \to \mathbb{R}$, $n = 1, 2, \cdots$ are measurable functions. Assume that for every $\varepsilon > 0$,

$$\mu\{x : |f_n(x) - f(x)| \geq \varepsilon\} \leq a_n(\varepsilon)$$

for some sequence $a_n(\varepsilon) \to 0$ as $n \to \infty$.

Show that there is a subsequence f_{n_k} so that $f_{n_k}(x) \to f(x)$ for all x outside a set of μ-measure 0.

> *Solution.*

3. Toward Lusin's theorem

Problem 3.1

(The finite-valued case of Lusin.) Let $s : [0, 1] \to \mathbb{R}$ be a measurable function taking only finitely many values. Prove that for every $\varepsilon > 0$ there is a compact set $K \subseteq [0, 1]$ such that

$$m([0, 1] \setminus K) < \varepsilon \quad \text{and} \quad s|_K \text{ is continuous.}$$

> *Solution.*

4. Optional qual problems

Problem P1

(Measurable solutions of Cauchy's equation; Bass 5.11.) Suppose $f : \mathbb{R} \to \mathbb{R}$ is Lebesgue measurable and

$$f(x + y) = f(x) + f(y) \quad (x, y \in \mathbb{R}).$$

Prove that $f(x) = cx$ for some $c \in \mathbb{R}$.

> *Solution.*

Problem P2

(UCLA Analysis Qual, Fall 2020, Problem 1.) Suppose $f : [0, 1] \times [0, \infty) \to [0, 1]$ is continuous. Prove that

$$F(x) = \limsup_{y \to \infty} f(x, y)$$

is Borel measurable.

> *Solution.*

Problem P3

(UCLA Analysis Qual, Fall 2011, Problem 1.) Let $f_n : [0,1] \to \mathbb{R}$ be measurable and suppose $f_n \to f$ almost everywhere. Prove that for every $\varepsilon > 0$ there is a measurable set $E \subseteq [0,1]$ with $m(E) < \varepsilon$ such that $f_n \to f$ uniformly on $[0,1] \setminus E$.

> *Solution.*